ITAYA0-44 Project

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Table of contents

1. Project
2. Instructions to Students
3. Section A
4. Terms of AI use for this Assessment
5. Question 1
6. How AI use Affects Marking

1. Project

Assignment / Project

Faculty:	Information Technology
Module Code:	ITAYA0-44
Module Name:	Data Analysis with Python
Content Writer:	Doreen Muderede
Internal Moderation:	Community of Practice
Copy Editor:	Phindile Tabata
Total Marks:	100
Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual assignment

This assignment contributes 30% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions to Students

Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted assignments.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your assignments.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

3. Section A

Section A

AI Use Requirements for This Assessment

This assessment is classified as AI-1 as defined in the AI Use Category Table below.

AI Use Category Table

For this specific assessment, the following applies:

Code	Category	What This Means	Evidence Required	Disclosure Requirement
AI-1	Limited AI Use	You may use AI only for simple support tasks (e.g. basic language and grammar, clarifying a definition or generating synonyms). The planning, brainstorming, research and final output must still be your own.	You will be required to attach full transcripts of all your AI chats for evidence of your process.	You must complete and submit an **AI Disclosure Appendix** explaining how you used AI. Access the AI Disclosure Appendix Template here.

4. Terms of AI use for this Assessment

Terms of AI use for this Assessment

1. What Counts as AI?

AI includes standalone or embedded generative systems that create or materially transform assessed work.

This includes AI features embedded in browsers, search engines, office applications, development environments, coding assistants, data platforms, security tools and engineering software.

The product name is less important than what the feature does.

See What Counts as AI

2. Permitted AI Activities

AI may suggest search terms, but it may not locate, select, summarise or evaluate sources for you.

See Permitted AI Activities

3. Assessed Concepts and General Terminology

You may not use AI to explain an assessed module concept as a substitute for engaging with the approved learning material. Each assessment will distinguish between:

Category	Meaning
Assessed concepts	Module knowledge, methods, technical processes or decisions that you must understand and demonstrate independently
General academic terminology	Terms such as peer review, Boolean operator, introduction or primary source that are not themselves being assessed
Borderline terminology	A term whose status is unclear and must be confirmed with the lecturer before using AI

4. Generally Prohibited Activities

These restrictions apply only where the activity forms part of the published assessment criteria. Assessment-specific instructions take precedence where they expressly permit or provide a tool, dataset, configuration, model, code component or other output. Please refer to the below and apply as appropriate.

See Generally Prohibited Activities

4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.

- 4.1.1. Software Engineering
- 4.1.2. Network Security Engineering
- 4.1.3. Data Science
- 4.1.4. Robotics

5. Question 1

Question 1 — 100 Marks

Study the scenario and complete the question(s) that follow(s):

Access to Basic Services and Service Delivery Protests in South Africa

You are a Junior Data Analyst working for the Department of Cooperative Governance and Traditional Affairs (CoGTA). Your manager has given you data on household access to basic services (water, electricity and sanitation) in South Africa and records of service delivery protests. Your task is to analyse, visualise, and present the results to a non-technical audience (e.g., ward councillors, community leaders, municipal officials).

You are required to:

- Pre-process the data in preparation for analysis
- Create effective graphs/charts that highlight key insights.
- Use storytelling techniques to present your findings logically and persuasively.
- Choose the right visualisation based on the audience and type of data.

Datasets Provided:

- Household Access to Basic Services in South Africa by Province (CSV)
- Service Delivery Protests in South Africa 2023 (Excel)

Write Python code to answer the following questions:

1.1 Loading & Inspecting Data (5 Marks)

a. Write Python code to load both datasets (`service_delivery_protests_sa_2023.xlsx` and `basic_services_access_sa.csv`) into pandas DataFrames. (1 Marks)
b. Display the first 5 rows of each dataset. (2 Marks)
c. Print the structure and data types of each dataset using appropriate pandas functions. (2 Marks)

1.2 Cleaning Data (10 Marks)

a. Write code to check for missing values in both datasets. (2 Marks)
b. Write code to remove or fill missing values appropriately. (3 Marks)
c. Identify and remove duplicate rows from both datasets. (3 Marks)
d. Write code to detect any unusual values (e.g., negative protest counts or access rates above 100%) (2 Marks)

1.3 Data Transformation (15 Marks)

a. Write code to standardise column names (convert to lowercase and replace spaces with underscores). (5 Marks)
b. Convert relevant columns (e.g., year, date) into proper datetime format. (5 Marks)
c. Encode categorical variables such as province or protest category using appropriate encoding techniques. (5 Marks)

1.4 Data Integration (15 Marks)

a. Write Python code to identify a common column (e.g., Province or Year) between the two datasets. (5 Marks)
b. Merge the two datasets into one DataFrame using an appropriate join method. (5 Marks)
c. Display the merged dataset and check for missing values after merging. (5 Marks)

1.5 Feature Engineering (15 Marks)

a. Create a new column that represents the total number of protests per province. (5 Marks)
b. Create a new column showing the average basic services access rate per province. (5 Marks)
c. Normalise or scale numerical columns using a suitable method. (5 Marks)

1.6 Data Visualisation (25 Marks)

a. Create a bar chart showing protest rates per province (e.g., protests per 100 000 households). (5 Marks)

b. Create a bar chart showing basic services access rates per province. (5 Marks)

c. Plot a line graph showing trends of protests over time. (5 Marks)

d. Plot a line graph showing basic services access trends over time. (5 Marks)

e. Create a scatter plot to show the relationship between basic services access rate and protest rate. (5 Marks)

1.7 Provide interpretation for the graphs in question 1.6 (15 Marks)

[Subtotal 100 Marks]

End of Question 1

6. How AI use Affects Marking

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment